Constrained Optimization

$$\min_{x \in \Omega} f(x)$$

f: smooth , Ω: $\{x \in \mathbb{R}^d \mid h(x) = 0\}$

Given a point x^*, how to check if it is a local optima.

x^* local optimal $\Leftrightarrow$ $\exists$ a small neighborhood $U \subset \Omega$ around x^*
s.t. $\forall\ x \in U$, $\quad f(x^*) \leq f(x)$

first order necessary condition for x^* to be a local minimizer ?

① $h(x^*) = 0$

② ?

Tangent cone of a convex set Ω.

What is a Tangent Vector

'd' is tangent to Ω at a point $x \in \Omega$. if there exist a sequence $\{z_k\}$ approaching x and a sequence $\{t_k\} > 0$ with $t_k \to 0$ such that

$$\lim_{k \to \infty} \frac{z_k - x}{t_k} = d.$$

$T_\Omega(x)$ is the collection of such tangent vectors
It is also called tangent cone.

Necessary condition for x^* to be locally optimal:

If x^* is a local solⁿ, then we have.

$\nabla f(x^*)^T d \geq 0 \quad \forall \; d$. s.t. d is a "tangent vector"

(is it a sufficient condition?)

Proof:

$$f(z_k) = f(x^*) + \langle \nabla f(x^*), z_k - x^* \rangle + o(\|z_k - x^*\|)$$

where $\lim_{r \downarrow 0} \frac{o(r)}{r} = 0$

$$= f(x^*) + t_k \nabla f(x^*)^T d + o(t_k)$$

(as $z_k = x^* + t_k d + o(t_k)$)

We assume $\nabla f(x^*)^T d < 0$ (Proof by contradiction), and x^* is local optima.

We get that

$$\frac{f(z_k) - f(x^*)}{t_k} = \underbrace{d^T \nabla f(x^*)} + \frac{o(t_k)}{t_k}.$$

eventually dominates for sufficiently large k.

$\therefore$ x^* is not a local solution

But this definition is more "geometric".

In order to check if x^* is locally optimal,

you need to find all sequences $\{z_k\}$ s.t $z_k \in \Omega$

and $z_k \to x^*$ and then get their limiting directions.

Difficult to do.

We want a more "algebraic" approach which can be codified for computing.

Quest for "algebraic" necessary conditions of local optimality.

$$\min_{x \in \Omega} f(x) \qquad \text{where. } \Omega \triangleq \{x \in \mathbb{R}^d \mid h(x) = 0\}$$

We will analyse the change in $f(x)$ and $h(x)$ as we take a small step 's' away from 'x'.

Let 's' be a small (but non-zero) step s.t.

$h(x+s) \approx 0$ (upto first order approximation)

$$\Rightarrow h(x+s) \approx \underbrace{h(x)}_{=0} + \langle \nabla h(x), s \rangle = 0$$

$\therefore$ s should satisfy $\langle \nabla h(x), s \rangle = 0$

for 's' to be feasible (upto first order)

Let s be a direction of descent from x
(upto first order approx)

$$f(x+s) = f(x) + \langle \nabla f(x), s \rangle$$

as $f(x+s) < f(x) \Rightarrow \langle \nabla f(x), s \rangle < 0$

$\therefore$ If x^* is locally optimal, it should satisfy the following:

For all first order feasible direction 's',

$$\langle \nabla f(x^*), s \rangle \geq 0$$

In other words,

$$\langle \nabla h(x^*), s \rangle = 0 \implies \langle \nabla f(x^*), s \rangle \geq 0$$

$$\forall s$$

How can this be written equivalently?

① Can we remove inequality from R.H.S?

② Can we then write: $\nabla f(x^*) = \lambda \nabla h(x^*)$

③ If x^* is not locally optimal, can we compute a direction of decrease?

from first order analysis at x^* (using first-order Taylor approximation)

we have the following:

① Construct a set $F_\Omega(x)$ of linearized feasible direction

$$F_\Omega(x) = \{ d \mid \nabla h(x)^T d = 0 \}$$

② If x^* is locally optimal sol^n, we get

$$\nabla f(x^*)^T d = 0 \quad \forall \; d \in F_\Omega(x^*)$$

In particular, we can write the first-order necessary condition for x^* to be a local optimizer as

$$\| \mathrm{Proj}_{F_\Omega(x^*)} (\nabla f(x^*)) \| = 0$$

However, all this analysis works only when

$F_\Omega(x) = T_\Omega(x)$.

One can always show that $T_\Omega(x) \subset F_\Omega(x)$

However, in order to show equivalence, we require some more conditions known as constraint qualification

One such condition is Linear Independence Constraint Qualification (LICQ). It holds when the active constraint gradients $\{ \nabla h_i(x) \}$ are linearly independent.

Others also exist e.g. constant rank constraint qualification.

So, when LICQ holds at $x \in \Omega$.

$$T_\Omega(x) = \{ d \mid \langle \nabla h_i(x), d \rangle = 0 \}$$

and first order necessary condition of optimality is

$$\| \mathrm{Proj}_{T_\Omega(x^*)} (\nabla f(x^*)) \| = 0$$